FACULTY OF COMPUTING

SEMESTER 2 2025/2026

SECP3213 – BUSINESS INTELLIGENCE

SECTION 1

PHASE 1 & 2: INTRODUCTION (PROPOSAL STAGE) & DATA WORKFLOW

PROJECT TITLE:

Business Intelligence Analysis of Brazilian E-Commerce Customer Purchasing Behavior Using Olist Datasets

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1.0 PROBLEM BACKGROUND

1.1 BUSINESS CONTEXT

With the rapid growth of the e-commerce industry, enterprise can now access a large number of transactions and customer data to optimize their operations. Olist, the main department store market for e-commence in Brazil are playing a crucial role by connecting thousands of small businesses across Brazil with customer. Operating in such large and competitive markets means tracking sales performance, understanding customer behaviour and monitoring logistics efficiency are absolute priorities. In order to convert a large amount of daily raw data in practical business insights, modern companies rely on business intelligence tools. Using Alteryx Designer for data preparation and Microsoft Power BI for analysis, organizations can make strategic, data-driven decision instead of relying on speculation.

1.2 IMPORTANCE OF THE PROBLEM

Although e-commence platforms generate a large amount of data every day, this information is often trapped in a separate silo such as isolated customer database, order logs, product details and shipping details. If there is no appropriate BI solution to connect these components, there will be several major operational problems.

First, Brazil's vast geographical environment makes transportation a headache, which are the difference between the expected delivery date and the actual delivery date often occurs and it will directly damage customer satisfaction. Secondly, if product details are not integrated with actual sales data, it is difficult o see which product categories really drive revenue and which product categories cause delivery delays or regional fulfilment problems. Finally, lack of customer segmentation means that the marketing team cannot identify high-spending groups, which lead to general activities that waste the budget. In order to prevent poor business forecasts and missed regional opportunities, it is obviously necessary to use Alteryx to clean up and model these data, laying a solid foundation for Power BI to track key performance indicators.

1.3 EXPECTED IMPACT

Building this BI solution will immediately improve market efficiency and revenue growth. By clearly planning delivery delays and shipping schedules in different states, management can reevaluate logistics partnerships and provide customers with more accurate delivery estimates. In terms of products, determining the best performing category and regional preferences will directly help manage inventory and attract suitable merchants to join the platform. In addition, dividing the customer group into clear consumption layers like low, medium, high and tracking their location will allow the marketing team to design highly targeted promotional activities to ensure that the budget is used for most profitable regions and customer segments.

2.0 OBJECTIVES

The main goal of this project is to build an end-to-end business intelligence solution that solve the data fragmentation issues within the Olist e-commence platform. To achieve this, the project will focus on these objectives:

1. To integrate and model four core e-commerce datasets (Customer, Order, Order Items and Products) from the Olist platform into a unified relational data structure using Alteryx.
2. To perform data cleaning and transformation processes, including handling missing values, removing duplicate records, standardizing data-time formats and creating custom analytical fields like spending_level.
3. To analyse retail sales trends, shipping fulfilment performance and product category revenue to identify meaningful business insights.
4. To segment the consumer base by geographic location and total purchasing value to provide a clear, structured framework for targeted marketing strategies.
5. To develop interactive dashboards using Microsoft Power BI for data visualization and business decision support.

3.0 DATASETS DESCRIPTION

This project uses four datasets from the **Brazilian E-Commerce Public Dataset by Olist** available on Kaggle.

Note on Dataset Changes

The original datasets selected for this project were changed due to difficulties in the data integration process. During the implementation stage, it was found that the datasets contained inconsistent fields and limited common keys, making the join process in Alteryx Designer difficult and inefficient.

As a result, the datasets were replaced with a new dataset that provides clearer relationships between tables and more suitable attributes for Business Intelligence analysis. The new dataset allows smoother data integration, transformation, and analysis processes while still meeting the project objectives and requirements.

3.1 DATASET 1 – olist_customers_dataset.csv

This dataset contains customer information and their geographic location data. It is structured data because it is provided in CSV format with rows and columns. This dataset is used to analyse customer distribution by city and state and understand regional purchasing patterns.

The key variables include:

Variable	Description	Variable	Description
customer_id	Unique customer identifier for each order	customer_city	Customer city name
customer_unique_id	Unique customer identifier	customer_state	Customer state name

3.2 DATASET 2 – olist_orders_dataset.csv

This dataset contains order-level information and order processing timestamps. It is structured data because it is provided in CSV format with rows and columns. This dataset is used to analyse order status, delivery performance, and transaction timelines.

The key variables include:

Variable	Description	Variable	Description
order_id	Unique order identifier	order_purchase_timestamp	Customer purchase timestamp
customer_id	Unique customer identifier for each order	order_delivered_customer_date	Actual order delivery date to customer
order_status	Order status such as delivered, shipped, and cancelled	order_estimated_delivery_date	Estimated order delivery date to customer

3.3 DATASET 3 - olist_order_items_dataset.csv

This dataset contains detailed information about products purchased within each order. It is structured data because it is provided in CSV format with rows and columns. This dataset is used to analyse sales performance, pricing, and shipping costs.

The key variables include:

Variable	Description	Variable	Description
order_id	Unique order identifier	price	Product price
product_id	Unique product identifier	freight_value	Product shipping cost
seller_id	Unique seller identifier		

3.4 DATASET 4 - olist_products_dataset.csv

This dataset contains product-related information and product category details. It is structured data because it is provided in CSV format with rows and columns. This dataset is used to analyse how product categories affect sales performance.

The key variables include:

Variable	Description	Variable	Description
product_id	Unique product identifier	product_category_name	Product category

4.0 PROPOSED BI APPROACH

4.1 TOOLS USED

The main tools used in this project are **Alteryx Designer** and **Microsoft Power BI.**

Alteryx Designer will be used for data preparation and workflow development. It used to import multiple Olist CSV datasets and do multiple functions to produce a cleaned dataset. The data preprocessing flows such as cleaning missing values, removing duplicates, transforming data types, joining related tables, creating calculated fields, and lastly exporting the cleaned dataset.

Microsoft Power BI will be used for data visualization, summarized report and dashboard development. It allows project team to create interactive dashboards, KPI cards, charts, slicers, filters, business reports and so on. Power BI is mostly suitable for this project because it can present business insights in a visual format that is easy for project team to understand and act as presentation use.

The combination of Alteryx and Power BI supports the full BI process from data preparation to insight generation. For the examples, Alteryx will handle the backend data processing while Power BI will focus on dashboard design and business storytelling such a win-win situation.

4.2 DATA INTEGRATION, CLEANING, AND TRANSFORMATION METHODS

The data preparation process will initiate by importing the selected datasets into **Alteryx Designer**. It will begin by importing four related Olist datasets such as orders, customers, order items, and products. For the example, the Olist dataset contains multiple related CSV datasets that can be joined using keys such as order_id, customer_id, and product_id.

Besides, data integration will be performed at an analytical level instead of treating them as one single database for the datasets from different source. The data analysis will compare common business dimensions such as sales, profit, customer segment, product category, region, purchase behaviour, and marketing response. This approach allows the project to build a meaningful retail BI solution without if all datasets belong to the same organization.

After that, the data cleaning process will also include removing duplicate records, handling missing values, correcting inconsistent data formats, standardizing column names, and removing unnecessary fields. For example, missing product categories will be replaced with "Unknown", and date fields will be converted into proper DateTime format. For instances, the missing product category names and incomplete delivery dates must be handled during data cleaning to ensure accuracy in product performance and customer spending segmentation analysis.

After doing the cleaning process, the data transformation process will be proceeded including forming new fields in datasets such as revenue, delivery_time_days, delivery_status, order_purchase_year_month, and spending_level. The cleaned and transformed data can be fully used for data analysis and dashboard visualization in Power BI.

After all data preparation process is completed successfully, the cleaned datasets will be exported from Alteryx and imported into **Power BI** for visualization. Now, **Power BI** will then be used to create dashboards that show business performance, customer insights, and marketing effectiveness.

4.3 PLANNED ANALYSIS

No.	Planned Analysis	Description	Expected Business Insight
1	Order Status Analysis	It is used to the number of orders based on order status.	Understand different order patterns.
2	Sales Trend Analysis	It is used to analyze monthly and yearly revenue using order purchase date.	Identify revenue growth patterns and sales trends over time.
3	Product Category Performance Analysis	It is used to analyze total revenue by product category.	Identify top product categories.
4	Regional Customer and Revenue Analysis	It is used to analyze customers and revenue by customer state.	Identify most customer regions and highest revenue states.

5	Delivery Performance Analysis	It is used to analyze delivery time and delivery status.	Evaluate performance and late deliveries.
6	Customer Spending Segmentation	Group customers into Low, Medium, and High spending levels.	Identify customer groups based on total spending behaviors.

4.4 PROPOSED VISUALIZATIONS

The project will use several visual forms in Power BI to present the analysis clearly and effectively.

Visual Form	Purpose
Line chart	It shows monthly or yearly sales trends.
Bar chart	It compares revenue by product category and customer state.
Map visualization	It displays customer distribution or revenue by state.
Donut chart	It shows order status distribution or customer spending segment distribution.
KPI Cards	It shows total revenue, total orders, total customers, and average delivery time.
Tree map	It shows product category contribution to total revenue.
Stacked column chart	It compares delivery status such as On Time and Late.

PHASE 2 (DATA WORKFLOW)

5.0 Data Workflow

This section explains the complete workflow using Alteryx Designer including data input, cleaning, transformation, integration, and analysis for the 4 datasets.

5.1 Data Input

Figure 5.1.1 shows the olist_orders_dataset.csv, olist_customers_dataset.csv, olist_order_items_dataset.csv, and olist_products_dataset.csv were imported into Alteryx Designer using the Input Data tool.

Figure 5.1.1 Data Input Workflow

5.2 Data Cleaning

Figure 5.2.1 shows the data cleaning was performed using Data Cleansing and Unique tools by removing null records, replacing missing numeric values with 0, replacing missing string values with blanks, and trimming whitespace.

For olist_orders_dataset.csv, duplicate records were removed based on the order_id field. For olist_customers_dataset.csv, duplicate records were removed based on the customer_id field. For olist_order_items_dataset.csv, duplicate records were removed based on the order_id and order_item_id field. For olist_products_dataset.csv, duplicate records were removed based on the product_id field.

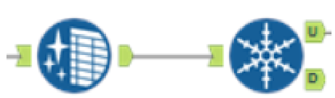


Figure 5.1.1.1 Data Cleaning Workflow

5.3 Data Transformation

Figure 5.3.1 shows data tranformation for olist_orders_dataset.csv was carried out using the Formula and Select tools. The DateTimeParse() function converted timestamp fields into DateTime format, while the Select tool standardized the datatypes of the transformed fields.

Figure 5.3.2 shows data tranformation for olist_customers_dataset.csv was carried out using the Select tools. The customer_zip_code_prefix field was deselected as it was not required for the analysis.

Figure 5.3.3 shows data tranformation for olist_order_items_dataset.csv was performed using the Select and Formula tools. The price and freight_value fields were converted into FixedDecimal data type, while the shipping_limit_date field was converted into DateTime data type. A new field named revenue was then created by summing the price and freight_value, and converted into FixedDecimal data type.

Figure 5.3.4 shows data tranformation for olist_products_dataset.csv was performed using the Select and Formula tools. All fields except product_id and product_category_name were removed as they were not required for the analysis. Lastly, null or empty values in the product_category_name field were replaced with "Unknown" using the Formula tool to improve data completeness and consistency.

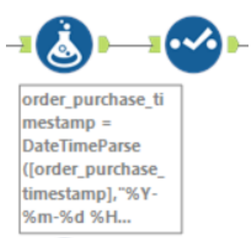


Figure 5.3.1 Data Transformation Workflow for olist_orders_dataset.csv

Figure 5.3.2 Data Transformation Workflow for olist_customers_dataset.csv

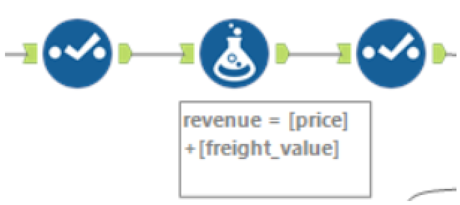


Figure 5.3.3 Data Transformation Workflow for olist_order_items_dataset.csv

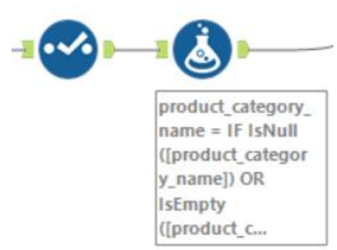


Figure 5.3.4 Data Transformation Workflow for olist_products_dataset.csv

5.4 Data Integration and Join Process

Figure 5.4.1 was performed between olist_orders_dataset.csv and olist_customers_dataset.csv using the customer_id. This join allowed each order record to relate to customer location details such as customer city and customer state.

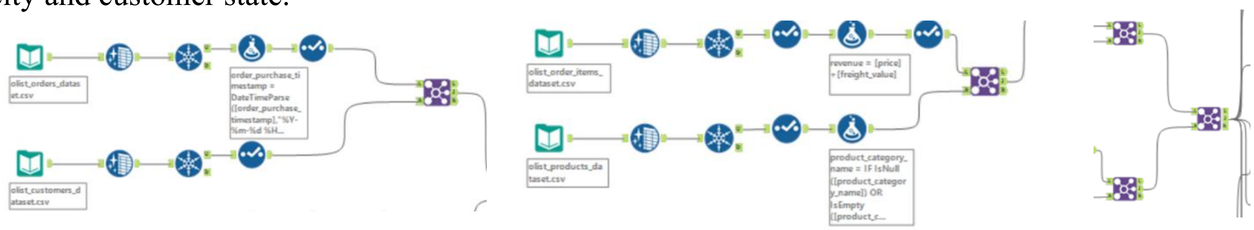


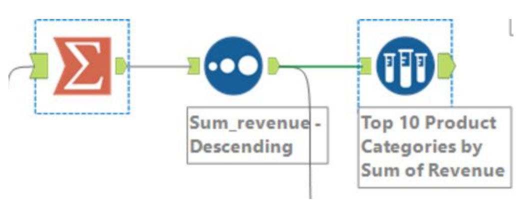
Figure 5.4.1 Data Join 1 & Figure 5.4.2 Data Join 2 & Figure 5.4.3Final Join

Figure 5.4.2 was joined between olist_order_items_dataset.csv and olist_products_dataset.csv using the product_id. Lastly, figure 5.4.3 was joined using the order_id to combine the order and customer information with the order item and product information. The final joined output provides the main data foundation for product category revenue analysis, regional revenue analysis, monthly sales trend analysis, and forecasting analysis.

5.5 Analytical Outputs

5.5.1 Product Category Revenue Analysis

Product Category Revenue Analysis was used to identify which product categories generated the highest total revenue in the Olist e-commerce dataset. This analysis was performed after the data integration process because some required fields came from different datasets.



Record	product_category_name	Sum_revenue
1	beleza_saude	1441248.07
2	relogios_presentes	1305541.61
3	cama_mesa_banho	1241681.72
4	esporte_lazer	1156656.48
5	informatica_acessorios	1059272.40
6	moveis_decoracao	902511.79
7	utilidades_domesticas	778397.77
8	cool_stuff	719329.95
9	automotivo	685384.32
10	ferramentas_jardim	584219.21

Figure 5.5.1 Product Category Revenue Analysis Workflow & Figure 5.5.2 Top 10 Product Categories by Total Revenue Output

In the Alteryx workflow, the dataset was first connected to a Summarize tool. The records were grouped and then a Sort tool was used to arrange the product categories in descending order based on total revenue.

A Sample tool was then used to display the Top 10 product categories by total revenue. The output of this analysis helps identify the strongest product categories in terms of sales contribution.

5.5.2 Regional Customer and Revenue Analysis

For revenue analysis, the Sample tool was used to display the Top 10 states with the highest total revenue. For customer analysis, another Summarize tool was used for group records and then the output was then sorted in descending order to identify the Top 10 states with the highest number of customers.

Figure 5.5.3 Regional Customer and Revenue Analysis Workflow

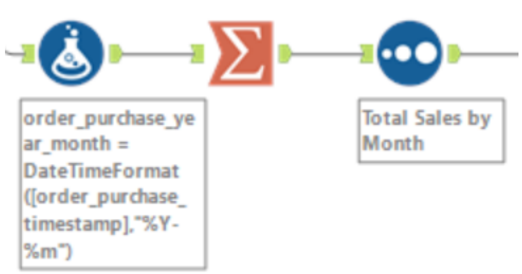
The results from this analysis help the business understand which regions contribute the most revenue and which states have the strongest customer base.

Record	customer_state	Sum_revenue
1	SP	5921678.12
2	RJ	2129681.98
3	MG	1856161.49
4	RS	885826.76
5	PR	800935.44
6	BA	611506.67
7	SC	610213.60
8	DF	353229.44
9	GO	347706.93
10	ES	324801.91

Record	customer_state	CountDistinct_customer_id
1	SP	39981
2	RJ	12303
3	MG	11178
4	RS	5249
5	PR	4840
6	SC	3513
7	BA	3257
8	DF	2062
9	ES	1956
10	GO	1942

Figure 5.5.4 Top 10 States with Highest Total Revenue Output & Figure 5.5.5 Top 10 States with Most Customers Output

5.5.3 Monthly Sales Trend Analysis



Record	order_purchase_year_month	Sum_revenue
1	2016-09	354.75
2	2016-10	56808.84
3	2016-12	19.62
4	2017-01	137188.49
5	2017-02	286280.62
6	2017-03	432048.59
7	2017-04	412422.24
8	2017-05	586190.95
9	2017-06	502963.04
10	2017-07	584971.62
11	2017-08	668204.60
12	2017-09	720398.91

Figure 5.5.6 Monthly Sales Trend Workflow & Figure 5.5.7 Total Sales by Month Output

The monthly revenue output was then sorted in ascending order based on the month field. The final output showed total revenue for each month and became the main input for both sales trend visualization and forecasting analysis. This analysis provides a clear view of revenue changes across months. It can help the business identify periods with stronger sales performance and periods with lower sales activity.

5.5.4 Forecasting Analysis Using ARIMA and ETS

Forecasting Analysis was conducted as an additional analytical technique to estimate future monthly revenue trends. First, a Formula tool was used to create a RevenueLog field using the LOG10 function. Next, another Formula tool was used to create a new field named month_date. After the date conversion, a Sort tool was used to arrange the records in ascending order by month_date. A Filter tool was then applied to keep only complete monthly records from January 2017 to August 2018.

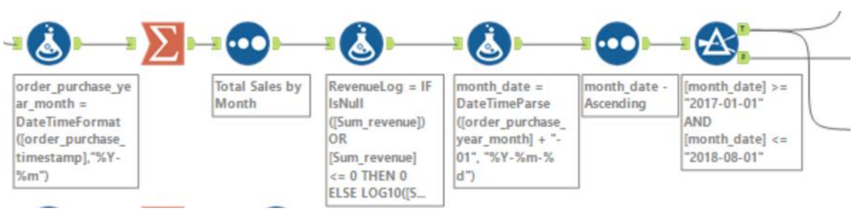
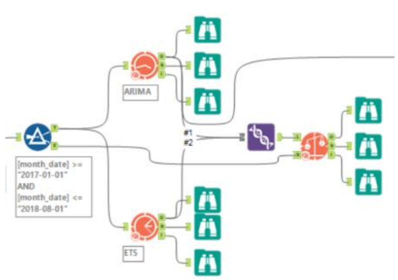
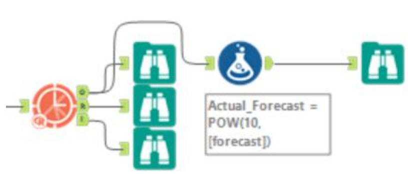


Figure 5.5.8 Forecasting Data Preparation Workflow

After the data preparation steps, both ARIMA and ETS models were applied and compared using the TS Compare tool. Based on the overall accuracy comparison, ARIMA was selected as the final forecasting model.



Record	Model	ME	RMSE	MAE	MPE	MAPE	MASE
1	ARIMA	-3.2837	3.5197	3.2837	-173.5687	173.5687	40.5128
2	ETS	-3.3487	3.5814	3.3487	-176.6078	176.6078	41.3148



Record	Period	Sub_Period	forecast	forecast_high_95	forecast_high_80	forecast_low_80	forecast_low_95	Actual_Forecast
1	2	9	6.000832	6.205895	6.134915	5.86675	5.79577	1001918.74
2	2	10	5.987199	6.324434	6.207705	5.766692	5.649963	970953.80
3	2	11	5.98361	6.529507	6.340553	5.626668	5.437714	962964.69
4	2	12	5.973046	6.728459	6.466984	5.479109	5.217634	939823.36
5	3	1	5.967326	6.972236	6.624401	5.310251	4.962417	927526.46
6	3	2	5.958242	7.224341	6.7861	5.130385	4.692144	908327.55

Figure 5.5.9 ARIMA and ETS Model Comparison Workflow & Result & Figure 5.5.10 Actual Forecast Workflow & Result

The TS Compare output showed that ARIMA achieved a MAPE of approximately 15.98%, while ETS achieved a MAPE of approximately 16.86%. Since ARIMA had the lower MAPE and better overall error performance, it was considered more suitable for the final forecast. The final Actual Forecast output showed a slightly decreasing future monthly revenue trend, with forecast revenue values ranging from approximately 1.02 million to 0.91 million. This indicates that the model predicted a moderate downward trend in future monthly revenue. However, since the historical monthly data range was limited, the forecast result was treated as a supporting analytical output rather than a fully conclusive prediction.

5.5.5 Delivery Performance Analysis

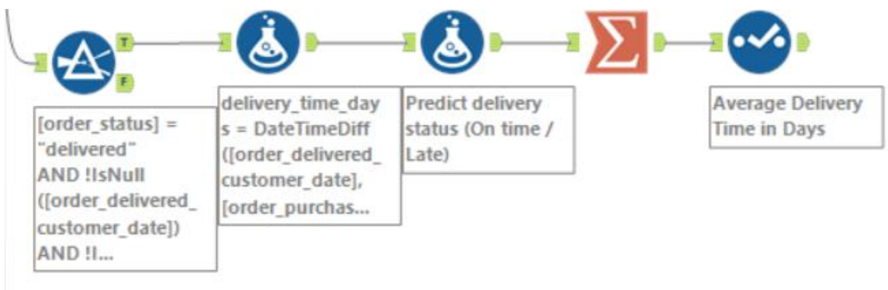


Figure 5.5.10 Delivery Performance Analysis Workflow

In order to get a better understanding of the logistics efficiency and customer satisfaction in the Olist market, customer orders are divided into two categories according to the estimated delivery time which are “on time” and “late”. As shown in Figure 5.5.11, 101475 orders were delivered within the expected time, with the average delivery time of 10.39 days, reflecting the overall stability and predictable logistics performance. In the other hand, 8714 orders were delayed and the average delivery time increased significantly to 30.89 days. This apparent gap indicates that when delays occur, they are usually driven by deeper supply chain disruptions within the regional delivery network rather than a slight scheduling deviation.

	delivery_status	Avg_delivery_time_days	Count
1	On time	10.386174	101475
2	Late	30.885931	8714

Figure 5.5.11 Distribution of Customer Orders by Delivery Status and Average Delivery Time

5.5.6 Customer Spending Segmentation

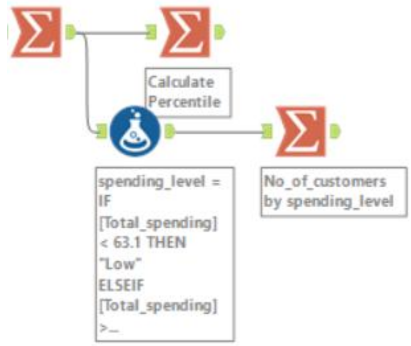


Figure 5.5.12 Customer Segmentation Workflow

To get a better understanding of customer purchasing behavior within the Olist marketplace, customers are divided into three spending levels which are low, medium and high based on the quartile analysis. As shown in figure 5.5.13, 23821 customers are classified as “Low” spenders, 47745 as “Medium” spenders and 23854 as “High” spenders. The distribution shows that most customers fall within the medium spending category while the remaining customers are evenly falling between the low and high spending category.

	spending_level	No_of_customers
1	Medium	47745
2	Low	23821
3	High	23854

Figure 5.5.13 Distribution of Customers by Spending Level

6.0 Challenges Faced

In the process of developing business intelligence solutions in Alteryx Designer, several technical challenges were encountered. One of the main challenges is data integration. Due to the lack of compatible keys in some tables, the originally selected dataset cannot be connected correctly. This led to incomplete connection and limits the effectiveness of the analysis process. In order to continue the project, our team migrated to the Olist e-commerce dataset, which provides a more structured relationship model for data integration.

Another challenge occurred in the data conversion stage when the output of the formula was accidentally assigned to the date field. This causes the original chronological information to be overwritten and affects the forecasting workflow. The problem was solved by storing the converted value in a separate field while retaining the original date column.

Our team also encountered difficulties in the prediction stage. Due to date anomalies and insufficient historical records, the initial prediction model produced unrealistic predictions. Therefore, the forecast does not accurately reflect the actual sales trend. In order to improve the performance of the model, the dataset is filtered to delete problematic records and keep a more consistent analysis time.

7.0 REFERENCES

1. Olist. (n.d.). *Brazilian e-commerce public dataset by Olist*. Kaggle. Retrieved May 11, 2026, from https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

Explanation (Why we suddenly change the phase 1)

1. During Phase 2 implementation, it was discovered that one of the datasets selected in Phase 1 did not contain a valid primary or foreign key, which prevented it from being properly joined with the other tables in Alteryx. To ensure data integrity and maintain a fully integrated workflow, the dataset was replaced with the official, well-structured Olist E-commerce Dataset.